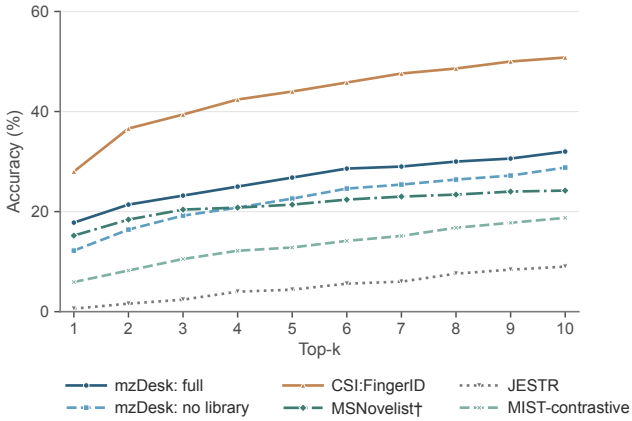
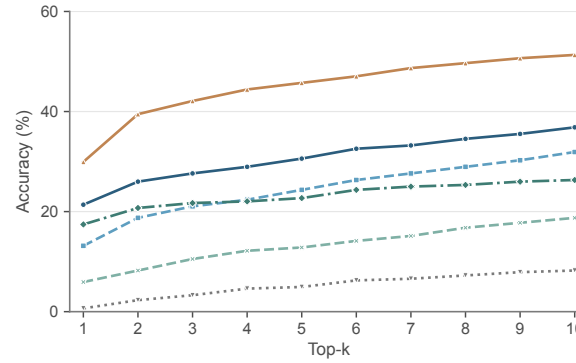
a Overall

n = 500; MIST: n = 304 (positive only)



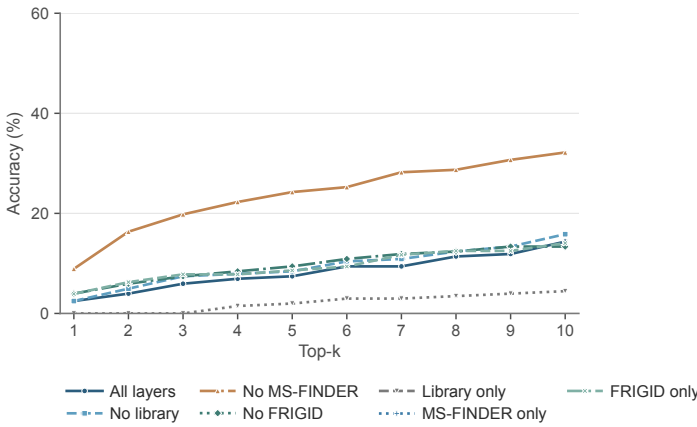
b Positive ions

n = 304



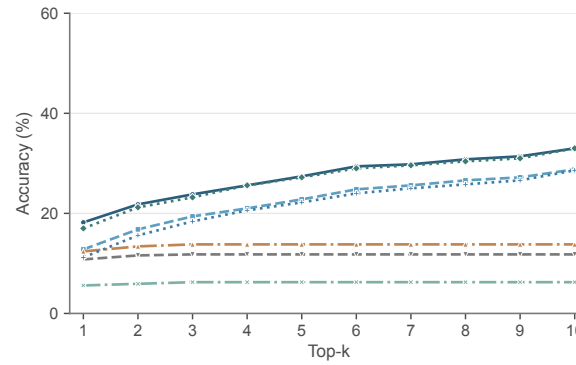
c Known-unseen subset

n = 202; MIST: n = 128 (positive only)



d Ablation

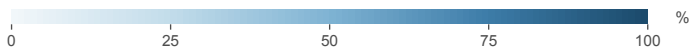
n = 500; FRIGID only: n = 304 (positive only)



e Overall statistics

Top-1 retrieval; n = 500; MIST: n = 304 (positive only)

	Precision	Recall	F1 score	Coverage
mzDesk: full	18.5	17.8	18.2	96.0
mzDesk: no library	12.8	12.2	12.5	95.4
CSI:FingerID	28.9	28.0	28.5	96.8
MSNovelist†	16.5	15.2	15.8	92.0
JESTR	0.6	0.6	0.6	96.6
MIST-contrastive	6.1	5.9	6.0	97.0



IK14 structure match; ties use the worst rank. No-output cases remain in the denominator.
Top-1 micro metrics: precision = correct / predicted; recall = correct / all cases;
F1 = 2 × precision × recall / (precision + recall); coverage = predicted / all cases.
MIST and FRIGID-only support positive ions only. Known-unseen excludes documented sets only;
† MSNovelist and FRIGID pretraining cannot be fully excluded. Candidate pools and settings vary.